Lecture Scribe 10

- Basics of Causal Inference:
    - Features t and outcomes y are utilized to build a model $p(y \mid t)$.
    - Causal inference seeks to determine if a change in t leads to a change in y.
- Challenges in Causal Inference:
    - It's impossible to observe all potential outcomes for a unit simultaneously—only the outcome under the actual treatment is observed, not the counterfactuals.
    - This problem highlights the inherent challenge in directly measuring causal effects.
    - Simply finding important features in regression doesn't confirm causality.
    - Other interventions post-feature collection can influence outcomes, prompting the need for a mathematical definition of causal inference.
    - Observational Challenges:
        - Directly comparing treated vs. untreated individuals to compute ATE can be misleading because treated individuals might inherently differ from untreated ones (sicker, older, etc.).
    - Importance of Randomization:
        - Treatment assignment must be independent of other variables to avoid bias and ensure the validity of causal estimates.
- Strategies for Overcoming the Fundamental Problem**:**
    - Emphasizes the importance of randomized experiments, where treatment assignment is random, to estimate causal effects.
    - Discusses statistical adjustments and assumptions (e.g., independence, homogeneity) that allow for causal inference from observational data.
- Solution Approach:
    - Observe the outcomes for both treatments identified by covariates x_i.
    - Build a model to predict potential outcomes and make predictions based on the model.
- Potential Outcomes Framework:
    - Potential outcomes are defined as the response if a particular treatment is given, for example,
    - Causal effects are functions of these potential outcomes, considering both the effect and the difference between treatments.
    - **The Rubin Causal Model:**
        - Introduces a formal model for causal inference that differentiates between potential outcomes under different treatments or interventions.
            - Learned in DS-GA-1002, potential outcomes contain the outcomes for all individuals under certain treatment conditions.

        - Want to investigate whether a drug is effective, PO1 denote the potential outcome of an individual that take the drugs, PO0 denote the potential outcome of an individual that don't take the drugs.
        - Compute the E(PO1) - E(PO0) and see whether it is positive.
            - If it is, then we say taking the drug causes the potential outcome(e.g. longevity) to increase.
    - Each unit (e.g., individual in a study) has potential outcomes that could be observed under each possible treatment, but only one outcome is observable since each unit can receive only one treatment.
        - The outcome not observed is called counterfactual.
- **Average Causal Effect (ACE):**
    - Defined as the average difference in outcomes between units under one treatment versus another, across the population.
    - I think it is central to understanding the overall impact of a treatment or intervention.
    - Can be related to Average Treatment Effect(ATE).
- **Assumptions for Causal Inference:**
    - **Independence:** The treatment assignment is independent of potential outcomes. p(PO)=p(PO|t)
        - This assumption only holds when the treatment is randomized or when we spot the confounding factor and adjust for that.
    - **Conditional Independence :** Conditioned on the confounding factor, there is no systematic difference between treatment and control group. p(PO|t,a) = p(PO|a)
    - **Homogeneity/Constant Effect:** The treatment effect is constant across units, simplifying the inference of causal effects.
- Healthcare Applications:
    - In healthcare, predictions on health status are crucial, often leading to questions on which medications or interventions to choose.
    - Causal inference is applied to decide which treatment might be more effective for a patient.